

BIT N BUILD 2026

- Problem Statement ID – 2
- Problem Statement Title – Debugging Silent Failures in AI Agent Executions
- Track – AI/ML
- PS Category – Software
- Project – Black Box: a flight recorder for AI agents
- GitHub repo:- https://github.com/OP141230/BlockForge_maharashtra_round

THE PROBLEM IN BRIEF

AI agents chain many tool calls. When the result is wrong, the failing step is rarely the visible one. Black Box records every step, locates the cause, and proves it by replaying from a checkpoint. Our AI identifies this problem and gives the user a clear answer as to where the AI agents working for him went wrong.

TECHNICAL APPROACH

Algorithmic Approach

Deterministic invariant checks plus small trained models, confirmed by checkpointed replay.

Tech Stack

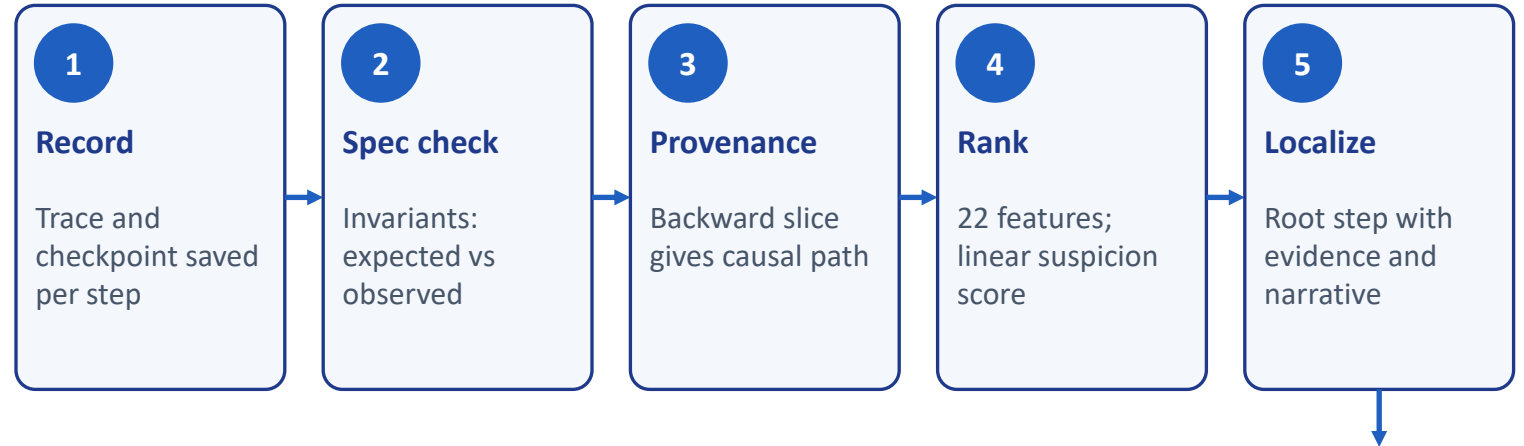
Python 3.10+

Standard-library core

scikit-learn

pandas

BLACK BOX DIAGNOSIS FLOW



Outputs: post-mortem narrative, contributions of each input/output, trace diff

WHAT EACH STEP LOGS (state checkpoint saved before and after every step)



TECHNICAL APPROACH

Learned Components

Step ranker

Linear scorer over 22 hand-built features, trained on 770 step examples (60 positive). Each score is weight $\times$ activation.

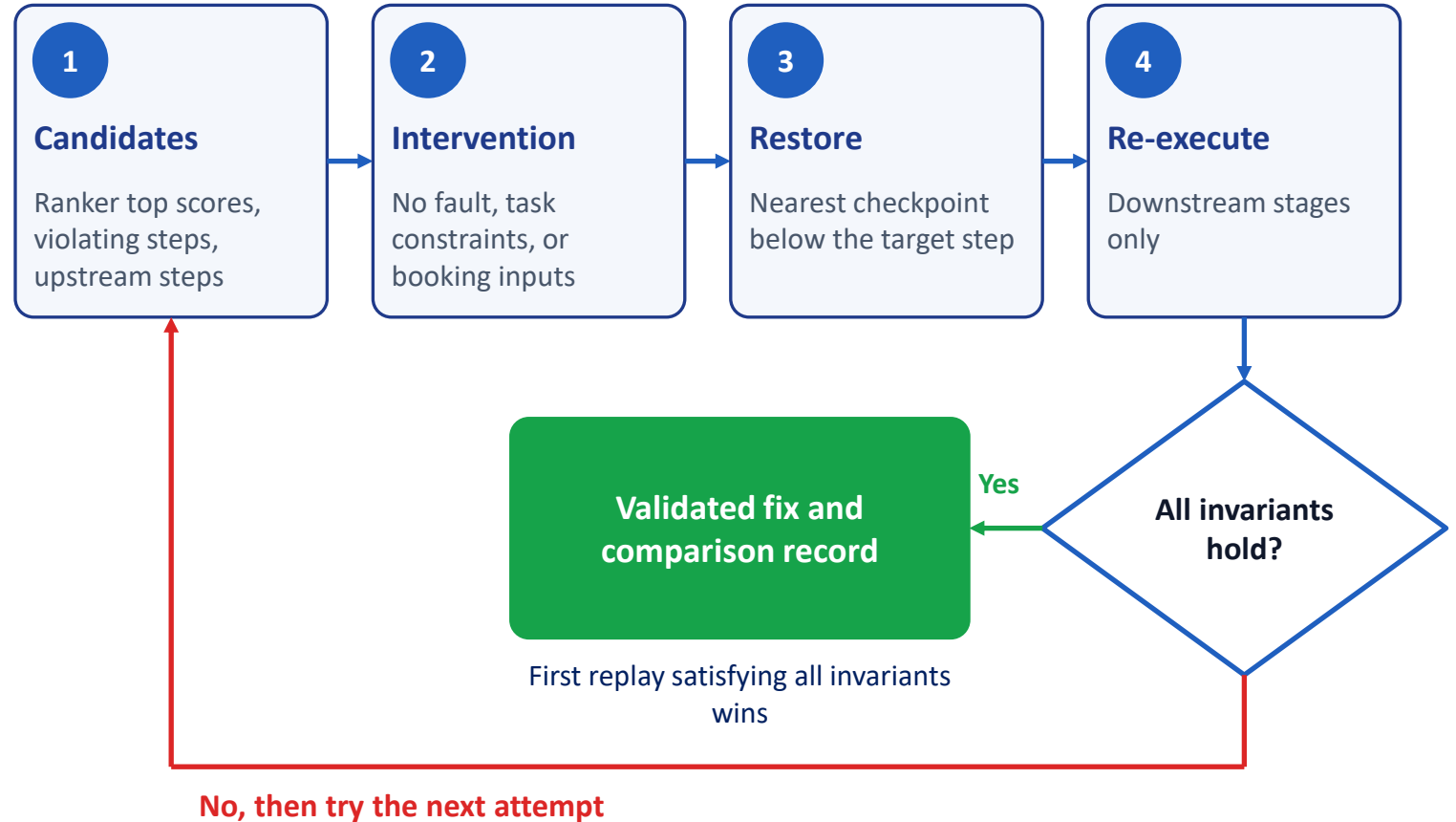
Intervention model

Logistic regression on 60 contrastive attempts (0.95 training accuracy). Annotates fix probability; ordering is off by default.

Trace comparison

Recursive diff of original vs replayed run: violations resolved, steps reused vs re-executed.

FLOW OF THE FAILURE IDENTIFICATION TECHNIQUE



FEASIBILITY AND VIABILITY

1. FEASIBILITY

Technical Feasibility:

Python 3.10+; the core pipeline uses only the standard library.

Operational Feasibility:

Every script validates its own output and exits non-zero, so the chain runs top to bottom.

Resource Feasibility:

No LLM calls; small linear and logistic models via scikit-learn.

2. ENHANCEMENTS & ADDITIONS

Real Tool Responses:

Replay recorded tool outputs so non-deterministic agents replay exactly.

Runtime Provenance:

Instrument state reads and writes instead of a hand-written map per agent.

Broader Validation:

Test the counterfactual search on more diverse agent traces.

3. RELIABILITY & EVIDENCE

Localization:

Top-1, Top-3 and MRR of 1.00 on 24 held-out failed traces.

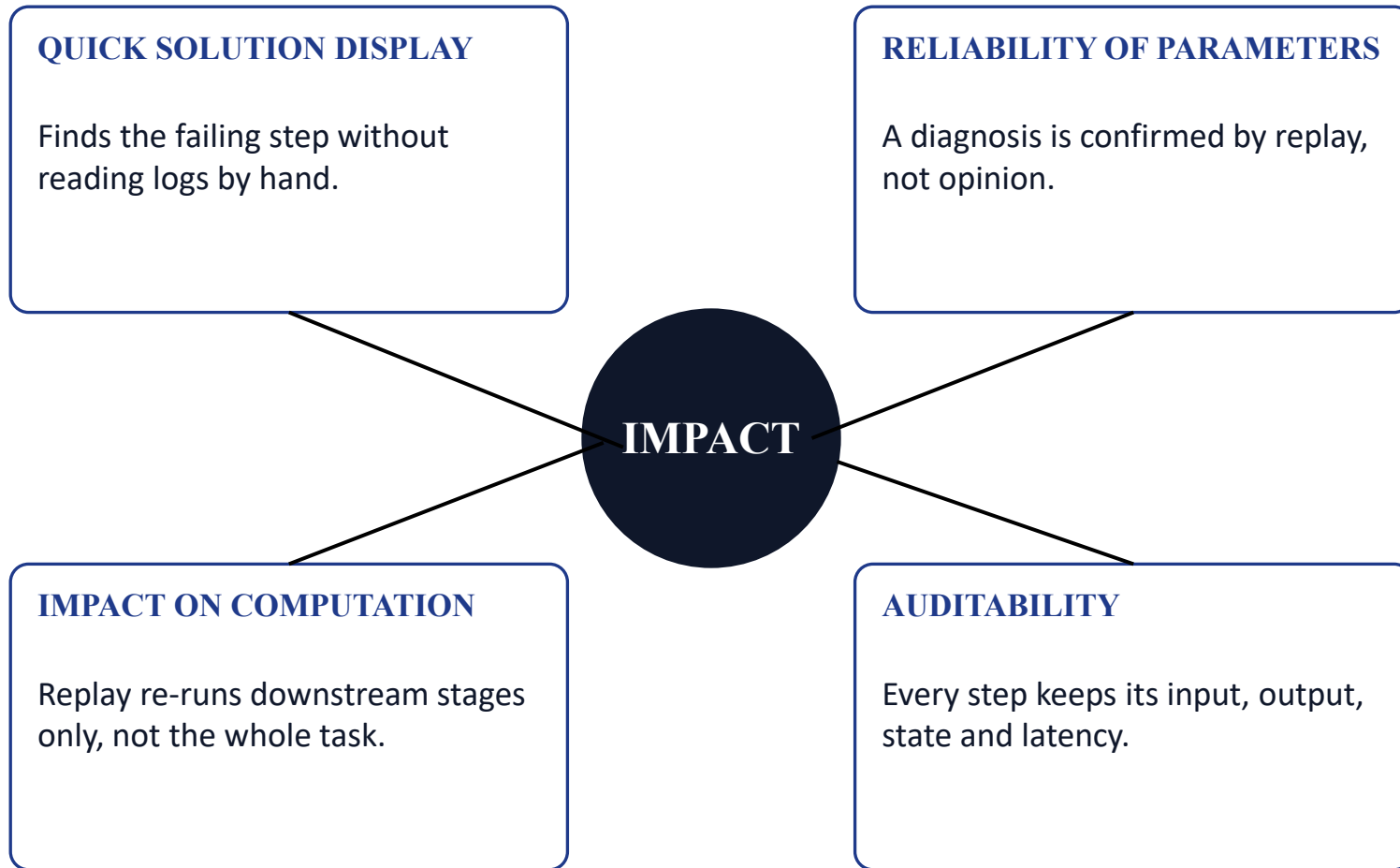
Replay Fixes:

6/6 known and 20/20 unseen-family traces fixed; 90/100 on a noisy bulk set.

Reproducibility:

All data is regenerated from the repo scripts.

IMPACT AND BENEFITS



KEY BENEFITS

- 1 EXPLAINABLE**
Linear ranker: every score is weight $\times$ activation.
- 2 GENERALIZES**
Fixed 20/20 traces from fault families never seen in training.
- 3 DETERMINISTIC**
No LLM in the diagnosis loop; results are reproducible.

Presented to you by –
Aditya Darke
Satyam Pawar
Om Patil
Vraj Dalal

THANK YOU